

ROYAL MANDI HOUSE

TECHNICAL REQUIREMENTS DOCUMENT

Engineering Specification — royalmandihouse.in

Project: royalmandihouse.in

Version: 1.0

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Prepared for: Royal Mandi House, Mangaluru

Status: APPROVED FOR DEVELOPMENT

1. Technology Stack

1.1 Core Technologies

Layer	Technology	Version	Justification
Markup	HTML5	Latest	Semantic HTML for SEO and accessibility
Styling	CSS3	Latest	Custom properties, Grid, Flexbox, animations
Scripting	Vanilla JavaScript	ES6+	No framework overhead, fast load, easy maintenance
Fonts	Google Fonts	CDN	Playfair Display (headings) + Inter (body)
Icons	Font Awesome	6.x CDN	Food, location, phone, social icons
Maps	Google Maps Embed API	v3	Free iframe embed, no API key required
Hosting	Netlify	Latest	Free tier, auto HTTPS, global CDN, drag-and-drop
Analytics	Google Analytics 4	Latest	Free, industry standard, Search Console integration

1.2 Why No Framework

- Vanilla HTML/CSS/JS loads 3–5x faster than React or Vue on mid-range Android devices
- No build step — developer edits file, refreshes browser, sees result immediately
- No node_modules, no version conflicts, no breaking changes from dependency updates
- AI tools (Claude, Cursor) can edit a single HTML file trivially
- Netlify deploys a single HTML file in under 30 seconds via drag and drop
- All required interactivity (tabs, slider, lightbox, animations) achievable in pure JS

2. File Architecture

2.1 Project Structure

The website will be built as a single HTML file with embedded CSS and JS, plus an assets folder:

File / Folder	Purpose
index.html	Main website file — all HTML structure, embedded CSS, embedded JS
blog-mandi-story.html	Blog post 1 — 'What is Mandi?'
blog-alfaham-vs-broasted.html	Blog post 2 — 'Al-Faham vs Broasted'
blog-mangalore-arabic-food.html	Blog post 3 — 'Mangalore and Arabic Food'
assets/images/logo.png	Restaurant logo — extracted from provided JPEG
assets/images/honey-chilly.jpg	Honey Chilly Chicken photo
assets/images/hero-1.jpg	Hero slide 1 — Arabic Mandi (Pexels)
assets/images/hero-2.jpg	Hero slide 2 — Biryani (Pexels)
assets/images/hero-3.jpg	Hero slide 3 — Broasted Chicken (Pexels)
assets/images/hero-4.jpg	Hero slide 4 — Kunafa (Pexels)
assets/images/gallery-1..8.jpg	Gallery images (Pexels)
sitemap.xml	XML sitemap for Google indexing
robots.txt	Crawler instructions — allow all
netlify.toml	Netlify configuration — cache headers, redirects

3. CSS Architecture

3.1 CSS Custom Properties (Design Tokens)

All colors, fonts, spacing, and transitions defined as CSS variables in :root for easy maintenance:

Variable	Value	Usage
--color-gold	#D4A017	Primary accent — buttons, borders, highlights
--color-gold-light	#E8B84B	Hover states on gold elements
--color-black	#0A0A0A	Primary background
--color-black-card	#111111	Card and section backgrounds
--color-white	#FFFFFF	Primary text on dark backgrounds
--color-text-muted	#AAAAAA	Secondary text, captions

Variable	Value	Usage
--font-heading	'Playfair Display', Georgia, serif	All headings H1–H4
--font-body	'Inter', Arial, sans-serif	Body text, nav, buttons
--transition-fast	0.2s ease	Button hovers, icon effects
--transition-med	0.4s ease	Tab switches, card hovers
--transition-slow	0.8s ease	Scroll animations, parallax
--border-radius	8px	Cards, buttons, inputs
--max-width	1200px	Content container max width
--section-padding	80px 0	Vertical section spacing (desktop)
--section-padding-mobile	50px 0	Vertical section spacing (mobile)

3.2 Responsive Breakpoints

Breakpoint	Width	Target Devices
Mobile (base)	320px – 767px	All Android and iOS phones
Tablet	768px – 1023px	iPad, large Android tablets
Desktop	1024px – 1439px	Laptops, desktops
Wide Desktop	1440px+	Large monitors, TVs

3.3 Grid System

- Navigation: Flexbox, space-between, align-center
- Hero: CSS Grid, single column, full viewport
- Highlights bar: CSS Grid, repeat(4, 1fr) desktop, repeat(2, 1fr) mobile
- Menu categories: CSS Grid, repeat(4, 1fr) desktop, repeat(2, 1fr) mobile
- Gallery: CSS Grid masonry via grid-template-rows: masonry or JS fallback
- Reviews: CSS Grid, repeat(3, 1fr) desktop, single column mobile
- Blog: CSS Grid, repeat(3, 1fr) desktop, single column mobile
- Contact: CSS Grid, 2 columns desktop (1fr 1fr), single column mobile

4. JavaScript Architecture

4.1 Module Breakdown

Module	Function	Implementation
HeroSlider	Auto-advances background images every 4s with crossfade	setInterval, CSS opacity transition
NavScroll	Shrinks navbar and highlights active section on scroll	IntersectionObserver + scroll event
MenuTabs	Switches menu category panels with sliding animation	data-tab attributes, CSS transforms
ScrollAnimations	Triggers fade-in-up on elements entering viewport	IntersectionObserver API
Parallax	Spotlight section background scrolls at 0.5x speed	scroll event, CSS transform translateY
GalleryLightbox	Opens full-size image on click, keyboard navigation	Dynamic DOM, keydown listener
StarAnimation	Fills gold stars when review section scrolls into view	IntersectionObserver, CSS animation
SmoothScroll	Smooth scroll to section on nav link click	scrollIntoView behavior smooth
MobileMenu	Opens/closes hamburger nav on mobile	classList toggle, aria-expanded
ButtonHover	Gold fill animation on all CTA buttons	CSS ::before pseudo-element, overflow hidden

4.2 Performance Rules

- No jQuery or any external JS library — all vanilla
- All JS at end of body (before </body>) to not block HTML parsing
- IntersectionObserver used for all scroll effects — no scroll event polling
- Images lazy loaded via loading='lazy' attribute on all non-hero images
- Hero images preloaded via <link rel='preload'> in <head>
- setInterval for slider uses requestAnimationFrame for smooth frames
- Event listeners use passive: true option where applicable

5. SEO Technical Implementation

5.1 HTML Head Tags

The following meta tags must be present in <head> of index.html:

Tag	Value
<title>	Royal Mandi House Best Mandi Restaurant in Mangalore — Hampankatta
meta description	Enjoy authentic Arabic Mandi, Biryani & Halal food at Royal Mandi House in Hampankatta, Mangalore. Open daily 12PM–12AM. Call +91 77954 16101.
meta keywords	mandi restaurant mangalore, arabic mandi mangalore, halal restaurant hampankatta, biryani mangalore, broasted chicken mangalore
meta robots	index, follow
canonical	https://royalmandihouse.in
og:title	Royal Mandi House — Dine Like Royalty in Mangalore
og:description	Mangalore's finest Arabic Mandi restaurant. Halal certified. Open daily 12PM–12AM.
og:image	https://royalmandihouse.in/assets/images/og-image.jpg
og:type	restaurant
viewport	width=device-width, initial-scale=1.0
theme-color	#D4A017

5.2 LocalBusiness Schema (JSON-LD)

The following structured data block must be embedded in <head> of index.html:

Schema Property	Value
@type	Restaurant
name	Royal Mandi House
address.streetAddress	Ground Floor, Beauty Plaza Building, Balmatta Rd, near Roopa Lodge
address.addressLocality	Hampankatta, Mangaluru
address.addressRegion	Karnataka
address.postalCode	575001
address.addressCountry	IN
telephone	+91 77954 16101
openingHours	Mo-Su 12:00-24:00
servesCuisine	Arabic, Indian, Chinese, Halal

Schema Property	Value
priceRange	₹₹
url	https://royalmandihouse.in
geo.latitude	12.8706 (approx from Maps pin)
geo.longitude	74.8441 (approx from Maps pin)
hasMap	https://maps.app.goo.gl/5GRFSYKUNB5YD7my8

6. Image Optimization

Image Type	Max Width	Max File Size	Format	Loading
Hero slides	1920px	150KB each	WebP with JPEG fallback	Preload first, lazy rest
Category cards	600px	80KB each	WebP	Lazy
Gallery	1200px	120KB each	WebP	Lazy
Blog thumbnails	600px	60KB each	WebP	Lazy
Logo	400px	30KB	PNG (transparent bg)	Eager
Spotlight	1920px	180KB	WebP with JPEG fallback	Eager

All images must have descriptive alt attributes following this pattern:

- Logo: 'Royal Mandi House logo — Best Mandi Restaurant Mangalore'
- Food images: '[Dish name] at Royal Mandi House, Hampankatta Mangaluru'
- Gallery: 'Royal Mandi House [dish name] — Halal restaurant Mangalore'

7. Animation Specifications

Animation	Trigger	CSS Property	Duration	Easing
Hero slide crossfade	setInterval 4000ms	opacity 0→1	800ms	ease-in-out
Navbar shrink	scroll > 80px	height, padding, font-size	300ms	ease

Animation	Trigger	CSS Property	Duration	Easing
Section fade-in-up	IntersectionObserver	opacity, translateY(30px→0)	600ms	ease-out
Staggered card entry	IntersectionObserver	opacity, translateY, delay per card	600ms + 100ms delay	ease-out
Menu tab slide	click	translateX	300ms	ease
Tab indicator	click	left (position)	300ms	ease
Button fill hover	hover	::before scaleX 0→1	300ms	ease
Button lift hover	hover	translateY(0→-2px)	200ms	ease
Card zoom hover	hover	scale(1→1.05)	300ms	ease
Parallax spotlight	scroll	translateY (0.5x rate)	0ms (live)	none
Gallery lightbox open	click	opacity, scale(0.9→1)	300ms	ease
Star fill	IntersectionObserver	color + scale	300ms + 100ms stagger	ease
Gold shimmer footer	page load	CSS @keyframes shimmer	2000ms	linear

All animations must respect the user's prefers-reduced-motion setting:

- @media (prefers-reduced-motion: reduce) — disable all transitions and animations
- Hero slider continues but without crossfade (instant switch)
- Scroll animations disabled — elements visible immediately

8. Netlify Deployment Configuration

8.1 netlify.toml

The following configuration must be in netlify.toml for performance and security:

Configuration	Value	Purpose
publish directory	/ (root)	Deploy entire project folder
Cache-Control: images	max-age=31536000, immutable	1-year cache for images

Configuration	Value	Purpose
Cache-Control: HTML	max-age=0, must-revalidate	Always fresh HTML
X-Frame-Options	SAMEORIGIN	Security header
X-Content-Type-Options	nosniff	Security header
Referrer-Policy	strict-origin-when-cross-origin	Privacy header
Redirect: www to non-www	301 redirect	Canonical URL enforcement

8.2 Domain Setup Steps

1. Purchase / confirm royalmandihouse.in domain is active
2. Log in to Netlify, go to Site Settings > Domain Management
3. Add custom domain: royalmandihouse.in
4. Netlify provides nameservers or CNAME record
5. Update DNS at domain registrar (GoDaddy / BigRock / Namecheap)
6. Wait 24–48 hours for DNS propagation
7. Netlify auto-provisions SSL certificate via Let's Encrypt
8. Verify HTTPS works at <https://royalmandihouse.in>

9. Post-Launch Technical Checklist

Task	Tool	Priority
Submit sitemap.xml to Google Search Console	Google Search Console	P0
Verify all pages indexed	Google Search Console	P0
Test mobile usability	Google Mobile-Friendly Test	P0
Run Lighthouse audit	Chrome DevTools	P0
Test all phone number links (tel:)	Manual on mobile device	P0
Test Google Maps embed loads	Manual	P0
Verify SSL certificate (HTTPS)	Browser padlock	P0
Set up Google Analytics 4	analytics.google.com	P1
Create Google Business Profile	business.google.com	P1
Test in Chrome, Safari, Firefox, Edge	Manual / BrowserStack	P1
Compress all images	Squoosh.app / TinyPNG	P1

Task	Tool	Priority
Add OG image for social sharing	Manual	P2

